

2023 Bi H2 Q1

Section: DNA and the Genome

Topic: Replication of DNA and PCR

Question Summary

At the start of DNA replication, the double helix unwinds and the two strands separate.

The question asks about:

- (a) the enzyme responsible for adding nucleotides,
- (b) why strand A is the lagging strand,
- (c) why two primers are used and what temperature they bind at,
- (d) a calculation of the number of DNA copies after several PCR cycles.

Worked Solution

(a) The enzyme that adds new DNA nucleotides is **DNA polymerase**.

It joins free nucleotides to the 3-prime end of a growing DNA strand using complementary base pairing.

(b) Strand **A** is the *lagging strand* because its synthesis is **discontinuous**.

DNA polymerase can only add nucleotides in the **5-prime to 3-prime** direction,

so strand A is made in short fragments
(*Okazaki fragments*) that are later joined together.

(c)(i) In PCR, **two different primers** are used — one for each end of the target DNA sequence.

Each primer matches a specific strand/end and ensures both complementary strands are copied.

(c)(ii) Primers bind to the target DNA at a temperature between **50 °C and 65 °C**.

This is the *annealing stage* of PCR.

(d) Each PCR cycle doubles the DNA.

9 minutes ÷ 3 minutes = 3 cycles → $30 \times 2^3 = 240$ copies.

Final Answers

Part	Answer	Marks
(a)	DNA polymerase	1
(b)	Lagging strand because made in fragments (5-prime to 3-prime direction only)	1
(c)(i)	One primer for each strand/end of the target sequence	1
(c)(ii)	50 °C – 65 °C	1
(d)	240	1

Revision Tips

- **DNA polymerase** adds nucleotides only from the 3-prime end (5-prime to 3-prime direction).
- **Primers** start replication — one per strand.
- PCR has three stages:
 1. **Denature** (95 °C)
 2. **Anneal** (50–65 °C)
 3. **Extend** (72 °C)
- Each PCR cycle **doubles** the DNA — 1, 2, 4, 8, 16, etc.